

SANTIAGO KLEIN

Evanston, IL, USA

☎ +1 773 800-5886

✉ santiagoklein@u.northwestern.edu

🌐 [santiago-klein](https://santiago-klein.github.io)

🐙 [sankle](https://github.com/sankle)

🌐 sankle.github.io

Education

Northwestern University

Ph.D. Student in Computer Science

Sep. 2024 – Expected Sep. 2029

Evanston, IL, USA

- Advised by Prof. Fabián E. Bustamante @ Aqualab.

Northwestern University

Master of Science in Computer Science (en route)

Sep. 2024 – Dec. 2025

Evanston, IL, USA

- GPA: 4.0

University of Buenos Aires (UBA)

Computer Engineering (Bachelor's + Master's equivalent)

Apr. 2017 – Oct. 2023

Buenos Aires, Argentina

- Specialization in Distributed Systems.
- GPA: 8.6/10. Graduated with honors.

Research Experience

AquaLab, Northwestern University

Research Assistant, advised by Prof. Fabián E. Bustamante

Sep. 2024 – Present

Evanston, IL, USA

- **Network Responses to Submarine Cable Disruptions:** Submarine cable failures routinely disrupt Internet connectivity, yet systematic methods for assessing network response remain limited. This project developed a measurement-driven approach based on large-scale traceroute data to characterize failure response strategies. By analyzing link-level path changes, the approach inferred whether traffic was rerouted locally or externally and revealed substantial variation in operator responses, ranging from seamless circuit-level protection to complete disconnection.

University of Buenos Aires (UBA)

Final Degree Project, advised by Dr. Esteban Carisimo

Mar. 2023 – Oct. 2023

Buenos Aires, Argentina

- **Submarine Platform Link Analysis System Hub (SPLASH):** Designed and implemented a network resilience analysis platform to quantify the impact of submarine cable failures on global Internet performance. Built an automated, high-throughput data pipeline integrating heterogeneous measurement sources (RIPE Atlas, MLab, PyPI), including ping, traceroute, and Root DNS CHAOS queries. Enabled fine-grained detection of traffic anomalies and latency degradation during infrastructure outages via custom Grafana dashboards.

Publications

- [1] Ten Years of the Venezuelan Crisis: An Internet Perspective. Esteban Carisimo, Rashna Kumar, Caleb J. Wang, **Santiago Klein**, Fabián E. Bustamante. *Proceedings of the ACM SIGCOMM 2024 Conference, Sydney, Australia*.

Work Experience

MuleSoft, a Salesforce Company

Software Engineering Member of Technical Staff (MTS; previously AMTS & Intern)

Apr. 2021 – Aug. 2024

Buenos Aires, Argentina

- Full Stack Software Engineer on the MuleSoft Anypoint Platform API Manager team, responsible for API lifecycle management, policy enforcement, and traffic control. Developed and scaled microservices handling ~1M req/s.
- Contributed to control-plane adaptations for managing the new Flex Gateway, and to the ground-zero deployment of all team-managed services on Hyperforce, Salesforce's public cloud infrastructure platform built on AWS.
- Technologies: JavaScript (React/Node), Java (Spring), PostgreSQL, AWS, Docker, Kubernetes, Istio, Redis, Jenkins, Spinnaker.

Teaching Experience

Northwestern University

Sep. 2025 – Dec. 2025

Teaching Assistant

Evanston, IL, USA

- CS340: Introduction to Networking. Topics: Internet architecture, TCP/IP, DNS, congestion control, routing and BGP, SDN, Ethernet and data center networks, mobility, network security.

University of Buenos Aires (UBA)

Jul. 2021 – Jul. 2022

Teaching Assistant

Buenos Aires, Argentina

- 75.41: Algorithms and Programming II. Topics: ADTs, algorithmic complexity, graphs.
- 75.08: Operating Systems. Topics: Kernel, processes, scheduling, memory management, advanced concurrency, file systems.

Projects

FIUBOOK

Feb. 2023

- University booking system enabling service providers to publish resources (e.g., books, classrooms, projectors) and users to discover and book them.
- Technologies: JavaScript (React/Node), Docker, Kubernetes, GraphQL.

Seedy FIUBA

Jul. 2021

- Blockchain-based crowdfunding application enabling transparent campaign funding and contribution tracking.
- Technologies: JavaScript (React/React Native/Node), Python, Solidity, Docker, Heroku, MongoDB, PostgreSQL.

Argentum Online

Jul. 2020

- Reimplementation of a classic MMORPG, adapted to run on Debian-based Linux systems.
- Technologies: C++, SDL2.

Skills

Programming Languages: C/C++, Rust, Go, Python, JavaScript, Java, SQL.

Systems & Data: Kafka, RabbitMQ, PostgreSQL, MongoDB, Redis, InfluxDB.

Infrastructure & Tools: Docker, Kubernetes, Terraform, AWS, Jenkins, Git, Grafana.

Languages: Spanish (Native), English (C2), French (B1).